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\documentclass[12pt,a4paper]{article}
% Fix for fancyhdr headheight error
\setlength{\headheight}{14.5pt}
% --- PACKAGES ---
% --- ENCODING & FONTS ---
\usepackage[utf8]{inputenc}
\usepackage[T1]{fontenc}
\usepackage{lmodern}      % Modern font (alternative to Times)
% \usepackage{times}      % OPTIONAL: Classic serif font (comment if using lmodern)

```

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\usepackage{pgf-pie}
\usepackage{pgfplots}
\usepackage{subcaption}
\usepackage{xcolor}
\usepackage{colortbl}
\usepackage{multirow}
\usepackage{booktabs}
\usepackage{siunitx}
\usepackage{float}

```

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% --- PAGE LAYOUT ---
\usepackage[a4paper, left=0.75in, right=0.75in, top=1in, bottom=1in]{geometry}
\usepackage{parskip}      % Paragraphs with spacing instead of indentation
\usepackage{setspace}     % For line spacing

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% --- MATHEMATICS ---
\usepackage{amsmath, amsfonts, amssymb}

```

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% --- GRAPHICS & FIGURES ---
\usepackage{graphicx}
\usepackage{caption, subcaption}
\usepackage{float}
\usepackage{booktabs}     % Better tables
\usepackage{array, longtable, colortbl}
\usepackage{multirow}     % Multirow cells in tables

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\begin{document}

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% --- MAIN CONTENT ---

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\section{Database Design}

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A robust and efficient database design serves as the backbone of \textbf{Chaldal.com's} e-commerce platform}, ensuring seamless data flow across departments—from inventory

management and order processing to customer support and delivery tracking. The proposed database system is designed to handle large volumes of transactions securely and efficiently while maintaining scalability to accommodate future business growth.

\subsection{Objectives}

The primary objective of the database design for \textbf{Chaldal Ltd.}, Bangladesh's largest online grocery delivery service, is to develop a \textbf{robust, efficient, and scalable data management system} that supports both operational and strategic needs. The design aims to integrate all functional domains—inventory, order processing, payment handling, customer service, and delivery management—into a unified and optimized data ecosystem, thereby ensuring smooth coordination and informed decision-making throughout the organization.

A well-structured and normalized database will not only enhance the performance and reliability of daily operations but also enable \textbf{data-driven insights} for long-term business growth.

\subsubsection{Specific Objectives}

\begin{enumerate}

\item \textbf{Efficient Data Organization and Storage:} Design a centralized data repository for structured storage and efficient retrieval of information related to customers, products, orders, inventory, and deliveries. This ensures coherence, accessibility, and consistency across all business units.

\item \textbf{Ensuring Data Integrity and Consistency:} Apply strong constraints, referential integrity, and validation mechanisms to maintain data accuracy and prevent anomalies, ensuring consistency across related entities.

\item \textbf{Data Security and Confidentiality:} Implement comprehensive security policies including data encryption, user authentication, and role-based access control to protect sensitive customer and company information from unauthorized access.

\item \textbf{Minimization of Data Redundancy:} Use normalization techniques to eliminate duplicate data, reduce storage overhead, and ensure that updates occur seamlessly without inconsistencies.

\item \textbf{Support for Multi-User Environment:} Develop a concurrent database system enabling multiple users—such as administrators, warehouse staff, delivery agents, and customer service personnel—to access and modify data simultaneously without performance degradation.

\item \textbf{System Scalability and Extensibility:} Ensure that the database can scale with business growth, accommodating an expanding product catalog, user base, and transaction volume, and adapting to new features or technologies.

\item \textbf{Enhanced Transaction Management:} Guarantee reliable transaction handling using ACID (Atomicity, Consistency, Isolation, Durability) properties to maintain data

reliability during operations such as order placement, payment processing, and inventory updates.

\item \textbf{Comprehensive Reporting and Analytical Capabilities:} Integrate interrelated datasets to facilitate the generation of dynamic analytical reports that support performance monitoring, trend prediction, and informed decision-making.

\item \textbf{Operational Efficiency and Process Automation:} Streamline workflows—from order placement to delivery—by automating routine processes, reducing manual intervention, and minimizing human errors.

\item \textbf{Improved Data Accessibility and Decision Support:} Provide timely and accurate access to critical information for both technical and managerial users to support efficient decision-making and optimized resource management.

\item \textbf{Customer-Centric Data Management:} Maintain comprehensive customer profiles including purchase history, preferences, feedback, and location details to enable personalized recommendations and improve customer satisfaction.

\item \textbf{Regulatory Compliance and Data Governance:} Ensure that the database complies with national and international privacy and data governance standards, maintaining transparency in data collection, storage, and processing.

\item \textbf{System Reliability and Fault Tolerance:} Design the database to ensure consistent performance with minimal downtime. Incorporate backup and recovery mechanisms to safeguard business continuity in case of data loss or system failure.

\end{enumerate}

\noindent\textbf{} By achieving these objectives, the proposed database system will serve as a comprehensive and reliable backbone for Chaldal Ltd. It will enhance operational efficiency, support data-driven decision-making, and improve overall user satisfaction. Furthermore, the design ensures adaptability to evolving market demands and technological advancements, reinforcing Chaldal's position as a leader in Bangladesh's online grocery delivery industry.

\subsection{Entity--Relationship Diagram}

\subsubsection*{Overview}

The database for \textbf{Chaldal Ltd.} integrates multiple entities that represent different operational components of the e-commerce grocery system. It manages information about customers, products, orders, payments, and deliveries in a relational and structured format.

Each entity is linked by key relationships that ensure data integrity and consistency throughout the business process—from product listing and order creation to delivery completion and payment confirmation.

\begin{figure}[H]

\centering

```
\includegraphics[width=1\textwidth]{er.png}
\caption{Entity–Relationship Diagram of Chaldal Ltd.}
\end{figure}
```

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\subsubsection*{Entities and Their Key Attributes}
```

```
\noindent\textbf{Customer}
\begin{itemize}
  \item \textbf{Customer\_ID (PK)}
  \item Name
  \item Email
  \item Mobile\_No
  \item Address
  \item Password
  \item Registration\_Date
\end{itemize}
```

```
\noindent\textbf{Category}
\begin{itemize}
  \item \textbf{Category\_ID (PK)}
  \item Category\_Name
\end{itemize}
```

```
\noindent\textbf{Product}
\begin{itemize}
  \item \textbf{Product\_ID (PK)}
  \item Product\_Name
  \item Price
  \item Description
  \item Perishability
  \item Availability
  \item \textbf{Category\_ID (FK)}
\end{itemize}
```

```
\noindent\textbf{Inventory}
\begin{itemize}
  \item \textbf{Inventory\_ID (PK)}
  \item \textbf{Product\_ID (FK)}
  \item Quantity
  \item Last\_Updated
\end{itemize}
```

```
\noindent\textbf{Order}
\begin{itemize}
  \item \textbf{Order\_ID (PK)}
  \item \textbf{Customer\_ID (FK)}
  \item Order\_Date
\end{itemize}
```

```

\item Status
\item Total\_Amount
\end{itemize}

```

```

\noindent\textbf{Order\_Item}
\begin{itemize}
\item \textbf{Order\_Item\_ID (PK)}
\item \textbf{Order\_ID (FK)}
\item \textbf{Product\_ID (FK)}
\item Quantity
\item Subtotal
\end{itemize}

```

```

\noindent\textbf{Payment}
\begin{itemize}
\item \textbf{Payment\_ID (PK)}
\item \textbf{Order\_ID (FK)}
\item Payment\_Method
\item Payment\_Status
\item Payment\_Date
\end{itemize}

```

```

\noindent\textbf{Delivery}
\begin{itemize}
\item \textbf{Delivery\_ID (PK)}
\item \textbf{Order\_ID (FK)}
\item Delivery\_Status
\item Delivery\_Agent
\item Delivery\_Date
\end{itemize}

```

\subsubsection*{Relationships Among Entities}

```

\begin{table}[H]
\centering
\begin{tabular}{||p{8cm}||c||}
\hline
\textbf{Relationship} & \textbf{Description} & \textbf{Type} \\
\hline
Customer $\rightarrow$ Order & A customer can have multiple orders, but each order belongs to one customer. & 1-to-Many \\
\hline
Category $\rightarrow$ Product & Each category can include multiple products. & 1-to-Many \\
\hline
Product $\rightarrow$ Inventory & Each product has one inventory record. & 1-to-1 \\
\hline
Order $\rightarrow$ Order\_Item & Each order can contain multiple products. & 1-to-Many
\end{tabular}

```

```

\hline
Product $\rightarrow$ Order\_Item & A product can appear in many orders. & 1-to-Many \\
\hline
Order $\rightarrow$ Payment & Each order has one payment transaction associated with it.
& 1-to-1 \\
\hline
Order $\rightarrow$ Delivery & Each order has one delivery record. & 1-to-1 \\
\hline
\end{tabular}
\caption{Relationships Among Entities in Chaldal Ltd. Database}
\end{table}

```

```

\begin{figure}[H]
\centering
\includegraphics[width =0.5\textwidth]{dbms.png}
\caption{Database of Chaldal}
\end{figure}

```

\subsubsection*{Text-Based Diagram Structure (Simplified Layout)}

%% Text-based representation of the ERD %%

\subsubsection*{Description of Entity Relationships}

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\begin{itemize}

```

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\item \textbf{Customer–Order Relationship:} Each customer can place multiple orders,
while each order must belong to one registered customer.

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\item \textbf{Order–Order\_Item Relationship:} Each order can contain multiple order
items; each item represents a specific product purchased within that order.

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\item \textbf{Order\_Item–Product Relationship:} Each order item corresponds to one
product, defining which product is included in the order and at what quantity.

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\item \textbf{Product–Category Relationship:} Products are classified under categories
such as \textit{Fruits, Vegetables, Baby Care, Staples}, etc., streamlining product
organization and filtering.

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\item \textbf{Product–Inventory Relationship:} Each product maintains an inventory record
that tracks its availability, quantity, and update timestamp.

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```

\item \textbf{Order–Payment Relationship:} Each order has a corresponding payment
record that includes method (e.g., cash, card, mobile payment) and status (e.g., pending,
paid).

```

```

\item \textbf{Order–Delivery Relationship:} Each order is associated with one delivery
transaction that records the delivery status, assigned agent, and completion date.

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\end{itemize}

```

\subsubsection*{Visual Representation Guide (Simplified ER Layout)}

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\centering
%% TikZ diagram of the ERD %%

```

```

\begin{figure}[H]

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\centering
\includegraphics[width=0.9\textwidth]{ers.png}
\caption{Entity–Relationship Diagram of Chaldal Ltd.}
\end{figure}

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\subsection{Database Table Design}

```

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\subsubsection*{4.3.3.1 Customer Table}

```

```

\begin{table}[H]
\centering
\begin{tabular}{|l|l|l|l|l|l|}
\hline
\textbf{Field Name} & \textbf{Key Type} & \textbf{Data Type} & \textbf{Size} & \textbf{Description} \\ \hline
Customer\_ID & Primary Key & VARCHAR & 10 & Unique identifier for each customer \\ \hline
Name & & VARCHAR & 50 & Full name of the customer \\ \hline
Email & & VARCHAR & 50 & Customer's email address \\ \hline
Mobile\_No & & VARCHAR & 15 & Contact number of the customer \\ \hline
Password & & VARCHAR & 255 & Encrypted customer password \\ \hline
Address & & VARCHAR & 255 & Physical or delivery address \\ \hline
Registration\_Date & & DATE & --- & Date when the customer registered \\ \hline
\end{tabular}
\caption{Customer Table Design}
\end{table}

```

```

\begin{figure}[H]
\centering
\includegraphics[width = 1\textwidth]{customer.png}
\caption{Customer Table}
\end{figure}

```

```

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\subsubsection*{4.3.3.2 Category Table}

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```

\begin{table}[H]
\centering
\begin{tabular}{|l|l|l|l|l|l|}
\hline
\textbf{Field Name} & \textbf{Key Type} & \textbf{Data Type} & \textbf{Size} & \textbf{Description} \\ \hline
Category\_ID & Primary Key & VARCHAR & 10 & Unique ID for each category \\ \hline
Category\_Name & & VARCHAR & 50 & Product category name (e.g., Fruits, Dairy, Household) \\ \hline
\end{tabular}
\caption{Category Table Design}

```

```
\end{table}
```

```
\begin{figure}[H]
\centering
\includegraphics[width =0.4\textwidth]{catagory.png}
\caption{Category Table}
\end{figure}
```

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```
\subsubsection*{4.3.3.3 Product Table}
```

```
\begin{table}[H]
\centering
\begin{tabular}{|l|l|l|l|p{6cm}|}
\hline
\textbf{Field Name} & \textbf{Key Type} & \textbf{Data Type} & \textbf{Size} & \textbf{Description} \\ \hline
Product\_ID & Primary Key & VARCHAR & 10 & Unique product identifier \\ \hline
Product\_Name & & VARCHAR & 100 & Name of the product \\ \hline
Price & & DECIMAL & (10,2) & Price of the product \\ \hline
Description & & TEXT & --- & Product description/details \\ \hline
Availability & & VARCHAR & 5 & Product stock status ("Yes/No") \\ \hline
Perishability & & VARCHAR & 5 & Product perishability ("Yes/No") \\ \hline
Category\_ID & Foreign Key & VARCHAR & 10 & Linked to Category Table \\ \hline
\end{tabular}
\caption{Product Table Design}
\end{table}
```

```
\begin{figure}[H]
\centering
\includegraphics[width =1\textwidth]{product.png}
\caption{Prodict Table}
\end{figure}
```

```
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```

```
\subsubsection*{4.3.3.4 Order Table}
```

```
\begin{table}[H]
\centering
\begin{tabular}{|l|l|l|l|p{6cm}|}
\hline
\textbf{Field Name} & \textbf{Key Type} & \textbf{Data Type} & \textbf{Size} & \textbf{Description} \\ \hline
Order\_ID & Primary Key & VARCHAR & 10 & Unique order identifier \\ \hline
\end{tabular}
```



```

Customer\_ID & Foreign Key & VARCHAR & 10 & Linked to Customer Table \\ \hline
Order\_Date & & DATETIME & --- & Date and time of order placement \\ \hline
Status & & VARCHAR & 20 & Order status ("Pending", "Delivered", "Cancelled", etc.) \\
\hline
Total\_Amount & & DECIMAL & (10,2) & Total cost of the order \\ \hline
\end{tabular}
\caption{Order Table Design}
\end{table}

```

```

\begin{figure}[H]
\centering
\includegraphics[width =1\textwidth]{order.png}
\caption{Order Table}
\end{figure}

```

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\subsubsection*{4.3.3.5 Order_Item Table}

```

\begin{table}[H]
\centering
\begin{tabular}{|l|l|l|l|l|p{6cm}|}
\hline
\textbf{Field Name} & \textbf{Key Type} & \textbf{Data Type} & \textbf{Size} & & \\
\textbf{Description} \\ \hline
Order\_Item\_ID & Primary Key & VARCHAR & 10 & Unique identifier for each order item \\ \hline
Order\_ID & Foreign Key & VARCHAR & 10 & Linked to Order Table \\ \hline
Product\_ID & Foreign Key & VARCHAR & 10 & Linked to Product Table \\ \hline
Quantity & & INT & 11 & Number of product units ordered \\ \hline
Subtotal & & DECIMAL & (10,2) & Total price for the item (Quantity  $\times$  Price) \\ \hline
\end{tabular}
\caption{Order\_Item Table Design}
\end{table}
\begin{figure}[H]
\centering
\includegraphics[width =1\textwidth]{order_item.png}
\caption{Order Item Table}
\end{figure}
%-----

```

\subsubsection*{4.3.3.6 Payment Table}

```

\begin{table}[H]
\centering
\begin{tabular}{|l|l|l|l|l|p{6cm}|}
\hline

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\textbf{Field Name} & \textbf{Key Type} & \textbf{Data Type} & \textbf{Size} &
\textbf{Description} \\ \hline
Payment\_ID & Primary Key & VARCHAR & 10 & Unique identifier for payment transaction \\
\hline
Order\_ID & Foreign Key & VARCHAR & 10 & Linked to Order Table \\ \hline
Payment\_Method & & VARCHAR & 20 & Method of payment ("Cash", "Card", "Mobile
Payment") \\ \hline
Payment\_Status & & VARCHAR & 20 & Status of payment ("Pending", "Paid", "Failed") \\
\hline
Payment\_Date & & DATETIME & --- & Date and time of transaction \\ \hline
\end{tabular}
\caption{Payment Table Design}
\end{table}

```

```

\begin{figure}[H]
\centering
\includegraphics[width =1\textwidth]{payment.png}
\caption{Order Table}
\end{figure}

```

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\subsubsection*{4.3.3.7 Delivery Table}

```

\begin{table}[H]
\centering
\begin{tabular}{|l|l|l|l|l|p{6cm}|}
\hline
\textbf{Field Name} & \textbf{Key Type} & \textbf{Data Type} & \textbf{Size} & &
\textbf{Description} \\ \hline
Delivery\_ID & Primary Key & VARCHAR & 10 & & Unique identifier for each delivery \\ \hline
Order\_ID & Foreign Key & VARCHAR & 10 & & Linked to Order Table \\ \hline
Delivery\_Status & & VARCHAR & 20 & & Status of the delivery ("Out for Delivery",
"Delivered") \\ \hline
Delivery\_Agent & & VARCHAR & 50 & & Name of assigned delivery personnel \\ \hline
Delivery\_Date & & DATETIME & --- & & Date and time of successful delivery \\ \hline
\end{tabular}
\caption{Delivery Table Design}
\end{table}

```

```

\begin{figure}[H]
\centering
\includegraphics[width =1\textwidth]{deliver.png}
\caption{Order Table}
\end{figure}

```

%-----

\noindent\textbf{Notes:}

\begin{itemize}

\item All primary keys are defined as \textbf{VARCHAR(10)} for consistency and scalability.

\item Foreign key constraints ensure \textbf{referential integrity} across dependent tables.

\item \textbf{DECIMAL(10,2)} is used for monetary values to maintain precision.

\item \textbf{VARCHAR} types are chosen for alphanumeric fields such as IDs, names, and categorical data.

\item \textbf{DATETIME} fields record important time-based events like order placement and delivery completion.

\end{itemize}

\end{document}